ESABU BLESSING OBEHI

Hardware Engineer

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Summary

Computer Engineering student with hands-on experience in embedded systems, FPGA development, sensor integration, and 3D mechanical design. Skilled in VHDL, C/C++, Arduino, and CAD, with project work spanning DSP, IoT health hardware, radar systems, and precision modelling. Active IEEE member with competition experience, seeking a hardware engineering internship.

Technical Skills

Core Competencies: Embedded Systems Design, FPGA Programming (VHDL), Circuit Analysis & Design, Sensor Integration & Interfacing, 3D Modelling & CAD, 3D Printing & Prototyping, Robotics & Control Systems, Digital Signal Processing, PCB Design Fundamentals.

Microcontrollers	Arduino, ESP32, Raspberry Pi
FPGA / HDL	DE10-Lite (Intel Altera); VHDL, Verilog
Languages	C, C++, Python, MATLAB
CAD / 3D	Autodesk Fusion 360, Autodesk Inventor, TinkerCAD, Bambu Studio
Sensors	MAX30102, Thermistor, HC-SR04 Ultrasonic, Servo

Experience & Community

Chip Design Trainee — **March 2026 – May 2026**

NITHub, University of Lagos — IEEE & NITHub Silicon Workshop — *Lagos, Nigeria*

- Hands-on digital IC design: RTL design in Verilog, synthesis, and the RTL-to-GDS flow, alongside lab instrumentation, PCB fundamentals, and embedded hardware in the NITHub makerspace.

Hardware Engineering Trainee — **2025 – Present**

BotsHub UNILAG — *Lagos, Nigeria*

- Student robotics and embedded-systems team; completed Arduino and TinkerCAD circuit-simulation training and built first two hardware projects through the programme.

Hardware Projects

Load Priority Controller (LPC) — Tiny Tapeout ASIC — Verilog · OpenLane · cocotb

- Designed an 8-load priority-based power manager in Verilog — configurable power budget (0–8), overload flagging, and registered outputs — currently being fabricated through the Tiny Tapeout open-silicon program.
- Verified with a 9-case cocotb testbench (all passing) and carried through the full RTL-to-GDS flow via OpenLane.

CardioTwin AI — Cardiovascular Screening Station — MAX30102 · Thermistor · ESP32 · Cloud

- Hardware interfacing lead (Team BitNomads): integrated the MAX30102 pulse oximeter and thermistor to capture HR, HRV, SpO2, and skin temperature from a fingertip station.
- Ensured reliable sensor-to-cloud data streaming under an end-to-end encrypted, NDPR-compliant architecture feeding a real-time CardioTwin risk score.

Twin Radar System — Object Detection & Visualization — Arduino · HC-SR04 · Servo · MATLAB

- Built the full hardware layer of a rotating radar: servo + ultrasonic sensor on Arduino with C firmware for sweep control and distance measurement, tracking objects across a 180° arc.
- Streamed live distance data to a MATLAB polar plot. **2nd Place**, UNILAG IEEE Week IES Hardware Competition (IEEE Nigerian Section).

Naijawatt Monitor — Smart Energy Usage Tracker — Arduino · Sensors · MATLAB

- IAS Girls team smart-energy monitor with predictive usage alerts; integrated sensing for power-consumption tracking with visualization and alert logic. Submitted as an IEEE IAS design project.

Moving Average Filter — FSM-Based DSP on FPGA — DE10-Lite · VHDL · Intel Altera

- Implemented a 3-tap running-average DSP filter as a finite state machine in VHDL on the DE10-Lite — onboard registers for delayed samples, switch input, and seven-segment output.

Helical Gearbox — 3D Mechanical Design Autodesk Inventor · Fusion 360 · Bambu Studio

- Designed and modelled the gear-head component for a team helical gearbox; 3D-printed the cover from Bambu Studio-sliced G-code into a functional assembly prototype.

CAD & Mechanical Design

- Modelled mechanical components in Inventor and Fusion 360 with assembly drawings and print-ready files — helical gearbox, bench vice, clamp, cable roller, caster wheels — exporting Bambu Studio G-code for FDM 3D printing.

Education

B.Sc. Computer Engineering **Expected 2029**

University of Lagos (UNILAG) | CGPA: 4.61/5.0 (First Class) *Lagos, Nigeria*

- Relevant coursework: Signals & Systems, Circuit Analysis (EEG 211/231), Introduction to Electrical & Electronics Engineering.

Competition & Recognition

- **2nd Place** — UNILAG IEEE Week IES Hardware Competition (IEEE Nigerian Section)
- **1st Place** — CJID Nigeria Hackathon
- **2nd Place** — Curacel Hackathon
- **3rd Runner-Up** — LCCI Hackathon

Certifications & Affiliations

- **Career Essentials in Generative AI** — Microsoft & LinkedIn
- **IEEE Member** — Robotics & Automation Society, Industrial Electronics Society, Industrial Applications Society (IAS-IPCSD design participant); IEEE & NITHub Silicon Workshop.

Leadership & Community

- **Founder**, Unilag Female Engineers '29 — community and mentorship for women in engineering at UNILAG.
- **Vice Secretary**, IEEE UNILAG Student Chapter — operations and administration.